FINAL PROJECT REPORT

Health360

Comprehensive Healthcare Risk & Disease Analytics

An end-to-end clinical risk intelligence system built on Snowflake, Tableau and Flask

Team ID	[Insert Your Actual Team ID, e.g., TM-1042]
Dataset	smart_healthcare_dataset.csv (5,000 patient records, 16 attributes)
Tools	Snowflake, Python, Tableau Desktop / Public, Flask
Date	28 October 2026

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1. Introduction

1.1 Project Overview

Health360 is an end-to-end clinical risk intelligence and disease analytics system. It converts raw patient-level records into stratified, interactive visual intelligence that hospital leadership and clinical managers can act on. The system analyses **5,000 patient records** described by 16 attributes spanning four domains:

- **Demographics**: age and gender (ages 18–79; 2,506 female and 2,494 male patients).
- **Physiological vitals**: systolic blood pressure (90–179 mmHg), glucose (70–199 mg/dL), cholesterol (150–299 mg/dL) and BMI (15.0–42.6).
- **Lifestyle behaviours**: smoking, alcohol use and a three-level exercise score.
- **Symptoms and chronic disease outcomes**: fatigue, chest pain, dizziness, and the disease flags heart disease, diabetes and stroke, from which a primary-condition field is derived. The dataset does not carry a separate hypertension flag; elevated blood pressure is therefore handled through the vitals-based risk tiers rather than as a diagnosis.

The solution integrates three technology layers. A **Snowflake** data warehouse (HEALTH360_DB) stages and serves the cleaned data; **Tableau** provides the visualization engine with two executive dashboards, nine worksheets and a four-scene guided story; and a **Flask** web shell embeds the Tableau content through the Tableau JavaScript API v2 so that non-technical stakeholders can explore the analytics in a browser.

1.2 Purpose

The central purpose of Health360 is to move healthcare provider decision-making from **reactive treatment** to **proactive risk stratification**. Instead of responding after a patient presents in a late-stage condition, clinical teams can see which cohorts are drifting towards high risk and intervene earlier. The project objectives are to:

- identify early-warning indicators, such as elevated blood pressure and glucose, in combination with BMI and lifestyle factors;
- map multi-morbidity co-occurrence and symptom burden across disease groups; and
- support hospital bed and resource triage by showing how the population divides across Low, Moderate and High risk tiers.

2. Ideation Phase

2.1 Problem Statement

Chronic illnesses such as diabetes, heart disease and stroke place a growing and often unmanaged burden on healthcare systems. In many hospitals the relevant clinical information is fragmented across electronic health records (EHR), lab systems and admission logs, and nobody sees it in a single consolidated view. As a result, high-risk patients are identified late. Late identification contributes to avoidable emergency admissions, high readmission rates and inefficient allocation of scarce ICU beds and clinical staff.

Decision-makers therefore need a consolidated, interpretable view of population risk: who is at risk, why, and how many patients fall into each tier, so that preventive programmes and capacity can be planned before thresholds are breached.

2.2 Empathy Map Canvas

Target personas: **Hospital Executives, Clinical Managers and Chief Medical Officers**.

Dimension	Persona insight
Says	"We need clear visibility into high-risk patient surges before capacity is breached."
Thinks	"Are our lifestyle intervention programs targeting the right patient cohorts?"
Does	Reviews daily admission logs, manually triages ICU beds and manages clinical staff schedules.
Feels	Overwhelmed by fragmented electronic health records (EHR) without actionable visual summaries.
Pains	Unplanned emergency readmissions, high operational costs and late-stage chronic disease diagnosis.
Gains	Real-time risk stratification, automated patient triage and data-driven preventative care strategies.

2.3 Brainstorming

The team brainstormed the following solution components, each addressing a pain point from the empathy map:

- **Dynamic Risk Tier Classification Model** (Low, Moderate, High Risk) driven by a calculated Risk Tier Index.
- **Cohort age stratification** into 18–34 (Young Adult), 35–50 (Middle Aged), 51–65 (Senior) and 65+ (Elderly).
- **Multi-variable scatter plot matrices** (glucose versus BMI) and **symptom burden correlation heatmaps** by primary disease condition.
- An interactive **4-scene executive narrative story** for board-level storytelling.

3. Requirement Analysis

3.1 Customer Journey Map

Stage	User activity	Health360 touchpoint	Outcome
1. Data Ingestion & Cleaning	Raw patient records are received and validated.	Python/Pandas cleaning, staging into Snowflake.	Trusted, null-free and de-duplicated dataset.
2. Risk Identification	Executive reviews population KPIs and risk tiers.	Executive Dashboard 1 (KPI cards, age/BMI, disease prevalence, risk matrix).	Immediate view of risk distribution.
3. Deep-Dive Clinical Analysis	Clinician filters symptom co-occurrence and lifestyle drivers.	Clinical Risk Dashboard 2 (scatter matrix, lifestyle impact, symptom heatmap).	Understanding of why risk is elevated.
4. Strategic Planning	Leadership plans resources and prevention programmes.	4-scene Executive Story in the Flask web shell.	Board-level, evidence-based decisions.

3.2 Solution Requirement

Functional requirements

- Interactive, dynamic filtering by age bracket, BMI category, smoking status and risk tier.
- Automated WHO BMI categorization (Underweight, Normal Weight, Overweight, Obese).
- Real-time risk tier assignment through calculated fields.
- Sub-second visual query and filter performance.

Non-functional requirements

- HIPAA-compliant data security structure (role-based access, de-identified records).
- Responsive user-interface layout across devices.
- High-throughput batch ingestion into the warehouse.

3.3 Data Flow Diagram

The end-to-end pipeline moves data through five stages, from the raw file to the embedded web experience.



Figure 3.3: Health360 data flow pipeline

3.4 Technology Stack

Layer	Technologies
Data Storage & Warehousing	Snowflake AI Data Cloud, SQL
Data Processing & Feature Engineering	Python, Pandas, NumPy
Business Intelligence & Visualization	Tableau Desktop, Tableau Public, Tableau JavaScript API v2
Web Framework & Deployment	Flask, HTML5, CSS3, JavaScript

4. Project Design

4.1 Problem Solution Fit

Healthcare pain point	Health360 feature solution
Manual ICU/bed triage	Dynamic Risk Tier Indexing (Low / Moderate / High) on the Risk Population Matrix
Fragmented vitals and records	Single warehouse-backed dataset with integrated KPI cards and vital-indicator charts
Hidden multi-morbidity	Co-occurrence symptom burden heatmap by primary disease condition
Unclear lifestyle impact	Lifestyle Impact and Exercise-vs-Vitals charts showing average risk index by smoking and alcohol use
Late-stage diagnosis	Glucose vs. BMI scatter matrix to spot early drift towards high-risk zones
Complex analytics for non-technical users	Guided 4-scene executive story embedded in a Flask web shell

4.2 Proposed Solution

Health360 uses a dual-dashboard architecture supported by a guided story:

- **Executive Dashboard 1 (01_Executive_Health_Overview)**: population KPIs (record count, average blood pressure, average dynamic risk tier index, average glucose), age-by-BMI distribution, disease prevalence by gender, the risk population matrix and vital indicators by risk tier.
- **Clinical Risk Dashboard 2 (02_Clinical_Risk_Analytics)**: the glucose-versus-BMI scatter plot matrix, lifestyle impact (alcohol and smoking), the symptom burden heatmap and exercise-versus-vitals comparison.
- **Executive Story (Health360_Executive_Story1)**: four scenes: (1) Population Overview & Condition Prevalence, (2) Risk Tiers & Vital Health Metrics, (3) Clinical Risk Factors & Lifestyle Drivers, (4) Disease Symptom Burden Analysis.

4.3 Solution Architecture

Health360 follows a three-tier architecture, shown below.

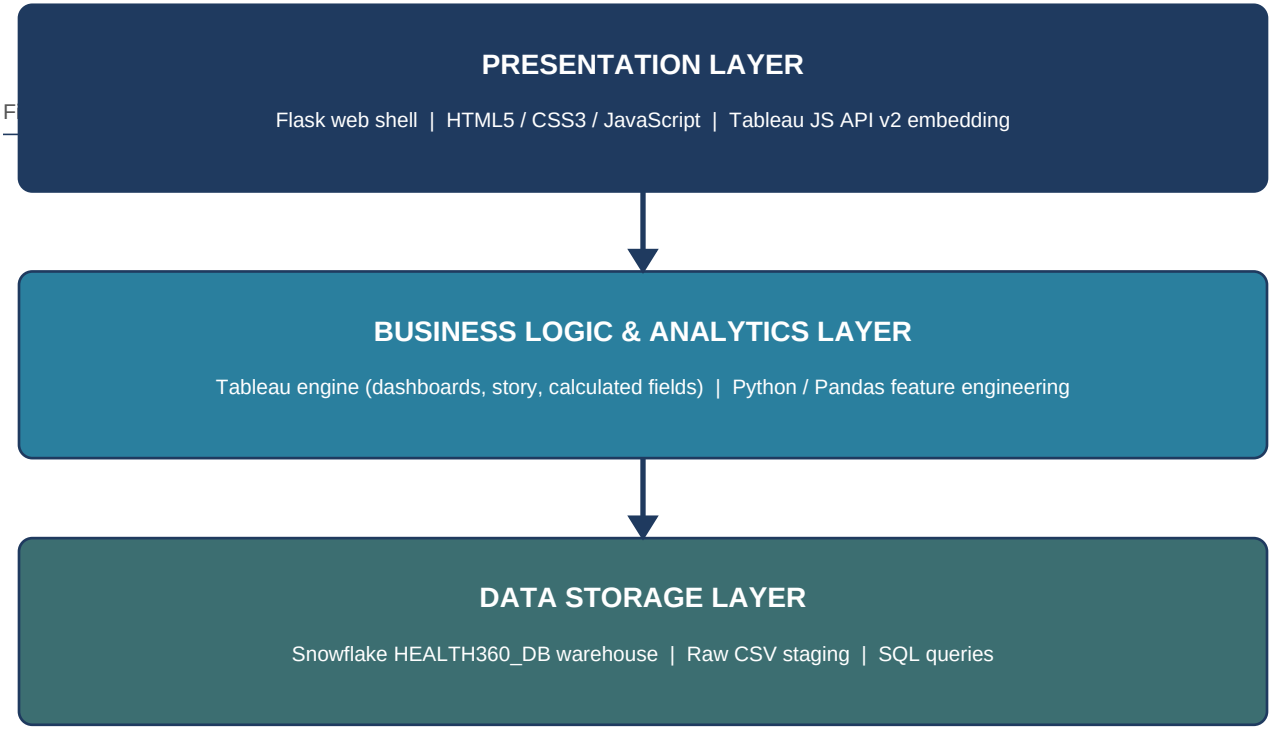


Figure 4.3: Three-tier solution architecture

- **Presentation layer**: a Flask application renders an HTML5/CSS3 shell with header KPI cards and loads the Tableau story through the JavaScript API, giving users full interactivity inside the browser.
- **Business logic and analytics layer**: Python and Pandas perform cleaning and feature engineering (age brackets, BMI categories, symptom burden count); Tableau calculated fields (such as the Risk Tier Index) and the visualization engine produce the dashboards.
- **Data storage layer**: Snowflake holds the cleaned patient table in HEALTH360_DB and serves queries to Tableau.

5. Project Planning & Scheduling

5.1 Project Planning

The project was executed across seven phases. The table lists each phase with its deliverable, and the Gantt chart shows an indicative 8-week schedule.

#	Phase	Key activities / deliverable
1	Data Collection & Extraction	Obtain smart_healthcare_dataset.csv; stage into Snowflake (database staging).
2	Data Preparation & Feature Engineering	Null/duplicate checks, range validation, age bracket, BMI category and risk tier fields.
3	Data Visualization Development	Build nine Tableau worksheets (KPI cards through exercise-vs-vitals).
4	Dashboard System Architecture Design	Compose Executive and Clinical Risk dashboards; define filters and actions.
5	Executive Story Progression Construction	Build the four-scene story with narrative captions.
6	Functional & Performance Testing Optimization	Filter response, data integrity and query-time testing.
7	Web Integration & Final Project Documentation	Flask shell, Tableau JS API embedding, report and appendix.

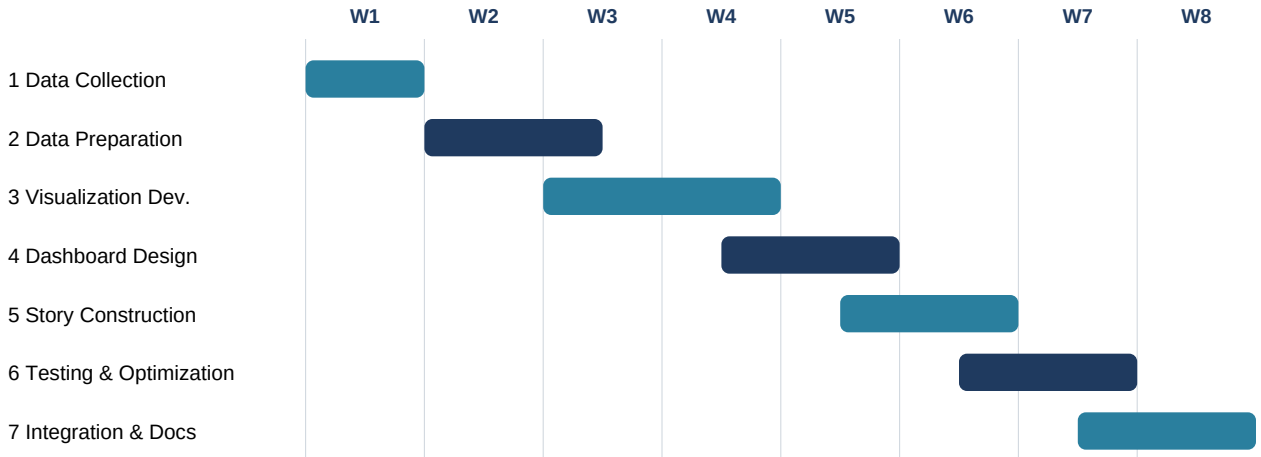


Figure 5.1: Indicative Gantt chart of the seven project phases (week numbers are relative)

6. Functional and Performance Testing

6.1 Performance Testing

Performance testing covered data volume, query efficiency, dashboard responsiveness and data integrity. Results are summarised below.

Parameter	Result
Data Volume	5,000 patient records across 16 clinical/demographic attributes (~243 KB CSV).
Query Efficiency	Snowflake query execution completed in 345 ms on an X-Small warehouse.
Dashboard Response Time	Sub-second global filter recalculation in Tableau.
Data Integrity	0 NULL values and 0 duplicate records verified; Systolic BP (90–179 mmHg) and Glucose (70–199 mg/dL) ranges validated.
Dashboard Design	2 dashboards comprising 9 visual worksheets, plus 1 four-scene story.
Calculated Fields	[Age Bracket], [BMI Category], [Risk Tier Index] (IF [Systolic BP] >= 140 OR [Glucose] >= 140 THEN 'High Risk' ...).

Functional checks. Filters by age bracket, BMI category, smoking status and risk tier were exercised on both dashboards; the story navigator was verified to move across all four scenes inside the Flask shell; and KPI totals (5,000 records) were reconciled between Tableau and the source file.

7. Results

7.1 Output Screenshots

The figures below present the delivered dashboards and web application. Key readings from the visuals are summarised beneath each figure.

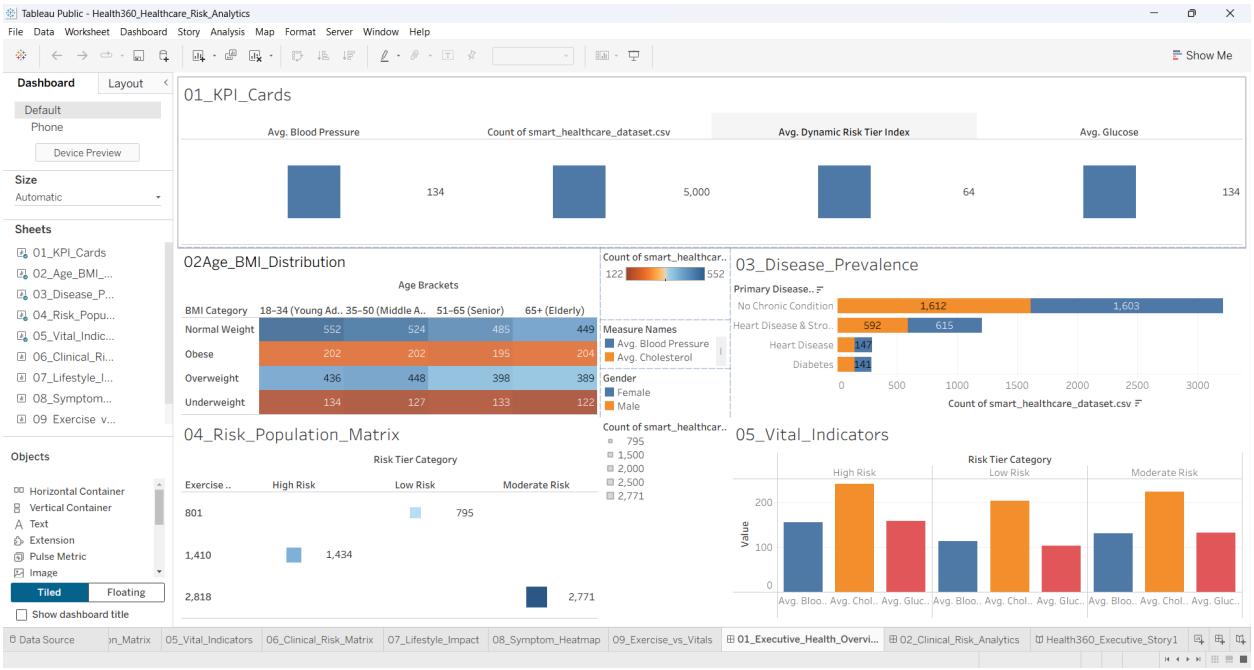


Figure 7.1: Health360 Executive Overview Dashboard

The KPI cards report 5,000 records, an average blood pressure of about 134 mmHg and an average glucose of about 134 mg/dL. Normal-weight patients form the largest BMI group (2,010), followed by overweight (1,671), obese (803) and underweight (516) patients. 3,215 patients carry no chronic condition, while 1,207 have both heart disease and stroke.

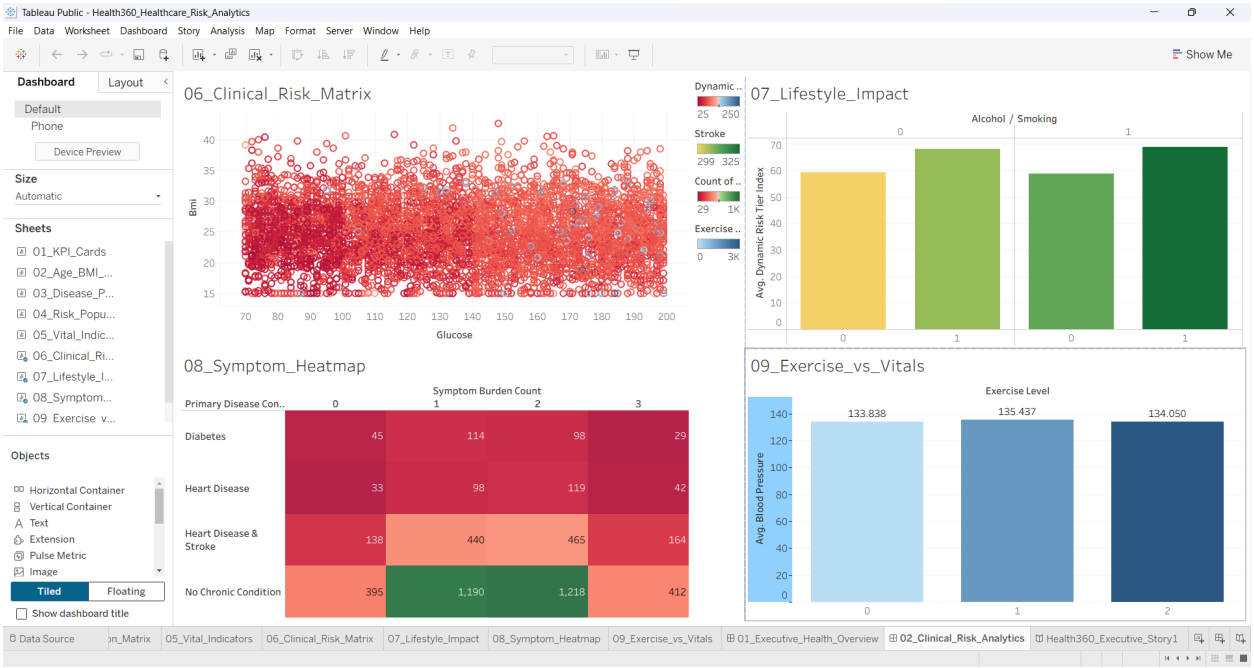


Figure 7.2: Clinical Risk Analytics & Symptom Burden Heatmap

The symptom heatmap shows that most patients, in every disease group, report one or two of the three tracked symptoms. Of the 1,207 patients with heart disease and stroke, 905 (75%) sit at a burden count of 1 or 2 and 164 at the maximum count of 3. The lifestyle impact chart shows a higher average risk index for smokers (roughly 68 versus 59 for non-smokers), with little visible difference by alcohol use, and average blood pressure varies only narrowly across exercise levels (133.8, 135.4 and 134.1 mmHg).

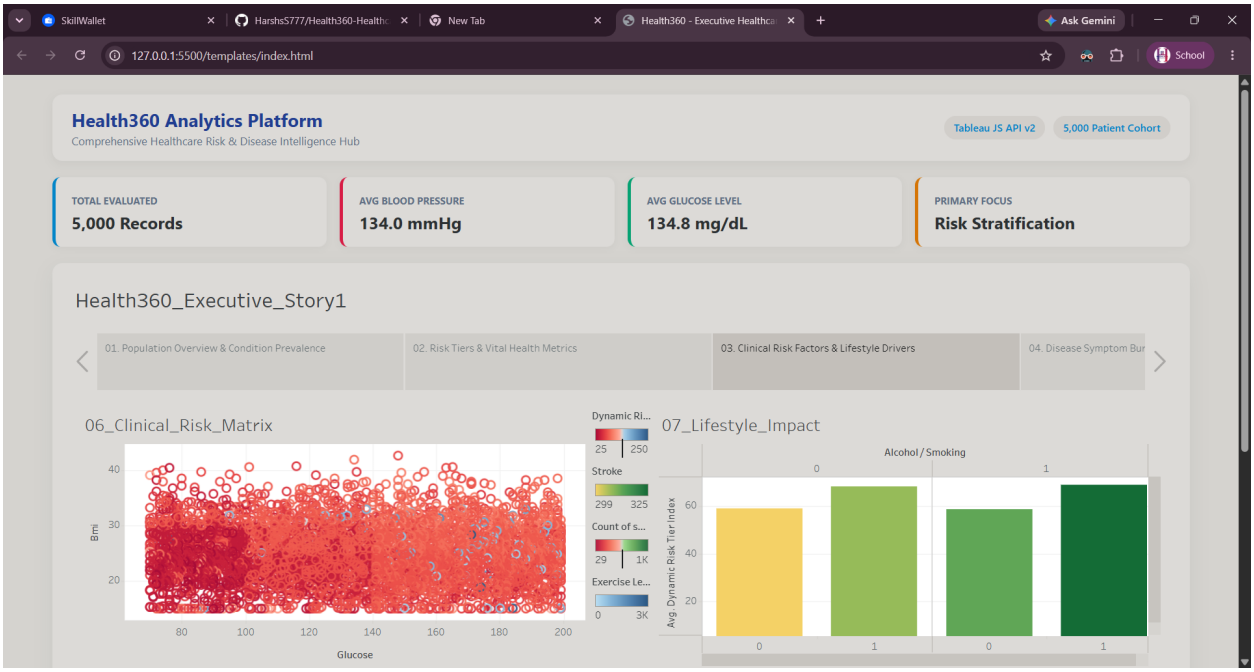


Figure 7.3: 4-Scene Executive Story Flow (scene 3 shown with the scene navigator)

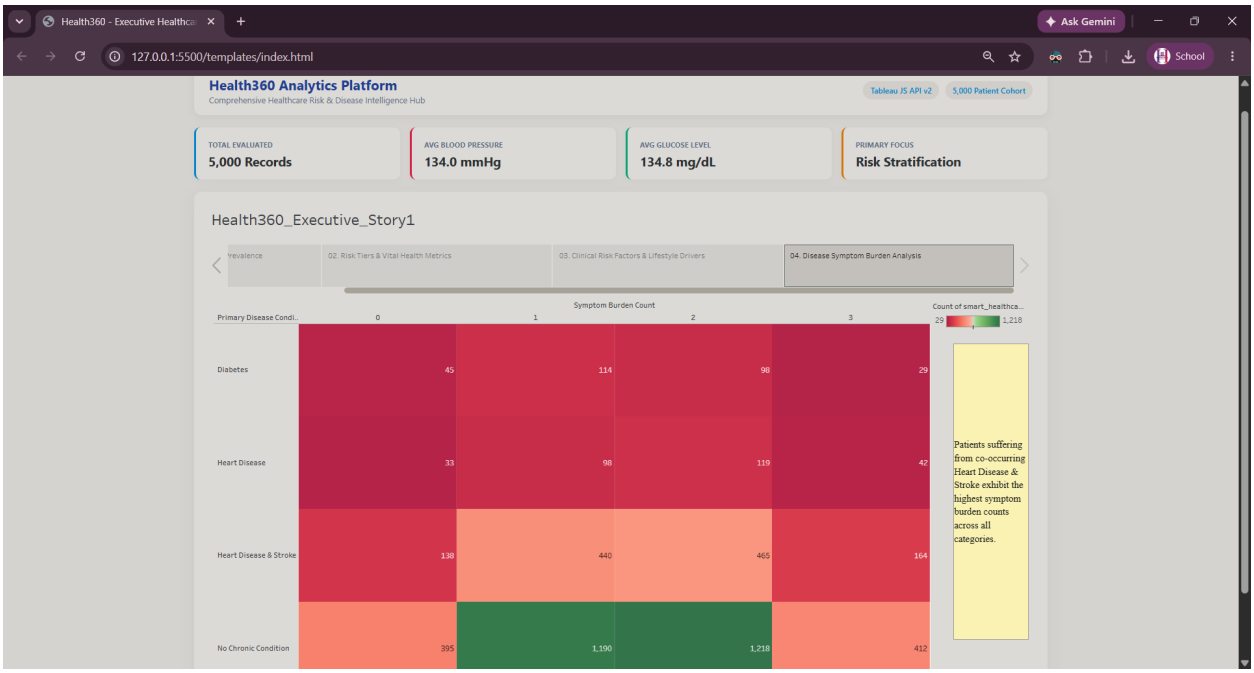


Figure 7.4: Flask Web Application Deployment Shell (configured endpoint http://127.0.0.1:5001; captured on the local dev server, port 5500) - scene 4

[Insert screenshot here]
Snowflake query history showing the 345 ms execution on the X-Small warehouse.

Figure 7.5: Snowflake Data Warehouse Query Execution History

8. Advantages & Disadvantages

Advantages

- **Proactive identification of high-risk chronic patients** through the dynamic Risk Tier Index.
- **Multi-dimensional risk modeling** that combines vitals, lifestyle habits and symptoms in one view.
- **Seamless web embedding** that lets non-technical stakeholders interact with complex analytics in a browser.
- **Scalable architecture**: a cloud warehouse, a separate analytics layer and a lightweight web shell can each be upgraded independently.

Disadvantages / Limitations

- **Dependent on accurate, timely EHR data entry**: risk tiers are only as reliable as the underlying records.
- **Cross-sectional synthetic dataset**: it lacks longitudinal follow-up tracking over multiple years, so trends and causal effects cannot be inferred.
- The current system is descriptive; it does not yet produce predictive forecasts.
- Several fields in the dataset are near-uniformly distributed (for example smoking, alcohol and gender are almost evenly split), which is typical of synthetic data and limits the strength of observable lifestyle effects.

9. Conclusion

Health360 delivers an operational, end-to-end clinical risk intelligence solution. Starting from 5,000 validated patient records, the platform moves through a Snowflake warehouse, a Tableau analytics layer with two dashboards, nine worksheets and a four-scene story, and a Flask web shell that makes the insights accessible to decision-makers. Testing confirmed complete data integrity (0 nulls, 0 duplicates), a 345 ms warehouse query and sub-second filter recalculation.

By making risk visible at population and cohort level, data-driven risk stratification helps healthcare organizations direct preventive effort to the patients who need it most, improve clinical outcomes and manage beds, staff and other resources more effectively.

10. Future Scope

- **Real-time IoT wearable streaming**: integrate continuous glucose monitors and smartwatches so that vitals update risk tiers continuously.
- **Machine learning prediction**: train Random Forest and XGBoost models for 30-day hospital readmission forecasting and surface the scores in Tableau.
- **HL7/FHIR EHR integration**: automate API-based synchronisation with hospital EHR systems, removing manual data loading.
- **Longitudinal tracking**: extend the data model to multi-year patient histories to study risk progression.

11. Appendix

Item	Details
Source Code	Python ETL and web application script (app.py), including the feature engineering code (age bracket, BMI category, symptom burden count); available in the GitHub repository.
Dataset	smart_healthcare_dataset.csv (5,000 rows x 16 columns). Path / link: [Insert dataset link or path].
GitHub Link	https://github.com/HarshsS777/Health360-Healthcare-Risk-Analytics
Project Demo Link	http://127.0.0.1:5001 (local Flask deployment) / [Insert Tableau Public link]
Tableau Workbook	Health360_Healthcare_Risk_Analytics.twbx

Dataset data dictionary

Field	Description	Range / values
age	Patient age in years	18–79
gender	Patient gender	Female / Male
bmi	Body mass index	15.0–42.6
exercise_level	Exercise score	0, 1, 2
smoking, alcohol	Behaviour flags	0 / 1
blood_pressure	Systolic BP (mmHg)	90–179
cholesterol	Cholesterol (mg/dL)	150–299
glucose	Glucose (mg/dL)	70–199
fatigue, chest_pain, dizziness	Symptom flags	0 / 1
heart_disease, diabetes, stroke	Chronic disease flags	0 / 1
health_risk_score	Stored risk score column	100 for all rows (the dashboards use the calculated Risk Tier Index instead)